VENDOR PERFORMANCE ANALYSIS

Inventory · Sales · Profitability Intelligence Report

Document Type	Records Analysed	Tools	Date
Analytics Report	**15.2 Million+**	**SQL · Python · Power BI**	**June 2026**

01 EXECUTIVE SUMMARY

This report presents a comprehensive analysis of vendor performance for a retail and wholesale distribution company. Raw transactional data spanning over 15.2 million records across four datasets was processed through a structured pipeline — from SQLite ingestion and SQL aggregation through Python-based statistical analysis to an interactive Power BI dashboard.

Nine business questions were addressed covering vendor dependency, brand profitability, bulk purchasing economics, inventory health, and statistical performance differences. The core finding — the Profitability Paradox — reveals that low-volume vendors consistently outperform high-revenue vendors on profit margin, a result confirmed statistically significant at p = 0.003.

$441.41M	**$134.07M**	**38.72%**	**$2.71M**	**65.7%**
Total Sales	Gross Profit	Profit Margin	Unsold Capital	Top-10 Vendor Share

02 PROJECT OVERVIEW

Business Problem

Effective inventory and sales management are critical for optimizing profitability in retail and wholesale. The company needed to address inefficient pricing, poor inventory turnover, and excessive vendor dependency. This analysis was structured around five core objectives:

- Identify underperforming brands requiring promotional or pricing adjustments
- Determine top vendors and brands contributing to sales and gross profit
- Quantify the impact of bulk purchasing on unit procurement costs
- Assess inventory turnover rates to reduce holding costs and improve efficiency
- Investigate the profitability variance between high and low performing vendors

Data Sources

Dataset	Records	Key Fields
purchase_prices.csv	12,261	Brand, Description, Price, Volume, PurchasePrice, VendorName
purchases.csv	2,372,474	VendorNumber, VendorName, Brand, PurchasePrice, Quantity, Dollars
sales.csv	12,825,363	Brand, SalesQuantity, SalesDollars, SalesPrice, ExciseTax, VendorNo
vendor_invoice.csv	5,543	VendorNumber, Quantity, Dollars, Freight, PODate, PayDate

Data Pipeline

Stage	Tool	Output
Data Ingestion	Python + SQLite + SQLAlchemy	4 raw CSV tables loaded into inventory.db
SQL Aggregation	CTEs + JOINs + GROUP BY	vendor_sales_summary — merged master table
Feature Engineering	Pandas + NumPy	GrossProfit, ProfitMargin, StockTurnover, SalestoPurchaseRatio
Exploratory Analysis	Matplotlib + Seaborn	Distribution plots, outlier detection, correlation heatmap
Statistical Analysis	SciPy	Two-sample T-test + 95% Confidence Intervals
Dashboard	Power BI Desktop + DAX	Interactive 9-visual report with 2 slicers

03 KEY FINDINGS & ANALYSIS

Finding 1 — Brands Requiring Promotional Intervention

Brands below the 15th percentile of sales but above the 85th percentile of profit margin were identified as high-priority promotional targets. These brands are profitable but under-exposed — carrying strong margins without the sales volume to realise their revenue potential.

Brand (Sample)	Total Sales	Avg Profit Margin	Status
Tequila Rose Strawberry	$312K	78.4%	Target Brand
Skrewball Peanut Butter Whiskey	$487K	74.2%	Target Brand
Casamigos Reposado Tequila	$621K	71.8%	Target Brand
St-Germain Elderflower Liqueur	$398K	69.5%	Target Brand
Hendrick's Gin	$544K	67.3%	Target Brand

Insight

High-margin, low-sales brands represent the highest ROI promotional opportunity in the portfolio. These products are profitable but invisible — they exist in the catalogue but fail to convert margin potential into revenue.

Recommended Action

Bundle target brands with top-5 selling SKUs in promotional offers. A conservative 20% sales uplift across these brands — with margin held constant — could add approximately $4.2M to gross revenue with zero new vendor investment.

Finding 2 — Sales Concentration in Top Vendors & Brands

Rank	Top Vendor	Revenue	% of Total
1	DIAGEO NORTH AMERICA INC	$68M	15.4%
2	MARTIGNETTI COMPANIES	$39M	8.8%
3	PERNOD RICARD USA	$32M	7.3%
4	JIM BEAM BRANDS CO.	$31M	7.0%
5	BACARDI USA INC.	$25M	5.7%

Recommended Action
Prioritise shelf space and contract renewals for DIAGEO NORTH AMERICA and MARTIGNETTI COMPANIES. Stockouts on Jack Daniel's or Tito's — the top two brands — could result in an estimated $150K–$200K weekly revenue loss.

Finding 3 — Vendor Procurement Concentration Risk

The top 10 vendors account for 65.7% of total purchase dollars. DIAGEO NORTH AMERICA INC alone represents 16.3% of all procurement — making any disruption to this single relationship a material operational risk.

Risk Assessment
At 65.7% procurement concentration, the business is highly exposed. A supply disruption or contract dispute with the top 3 vendors could reduce available inventory by over 40% within 60 days.

Recommended Action
Diversify the vendor base by onboarding 3–5 new qualified vendors for high-volume categories. Target: reduce top-10 procurement share from 65.7% to under 55% within 12 months.

Finding 4 — Bulk Purchasing Reduces Unit Cost by 72%

Order Size	Avg. Unit Purchase Price	Saving vs. Small
Small (< 200 units)	~$38.50	—
Medium (200–1,000 units)	~$22.10	43% lower
Large (> 1,000 units)	~$10.78	72% lower

Recommended Action
Shift fast-moving SKUs to large-order thresholds. Estimated annual procurement savings of ~$2.1M assuming 30% of current small/medium orders migrate to large tier. Avoid bulk purchasing of slow-moving items — holding cost will offset savings.

Finding 5 — $2.71M Capital Locked in Unsold Inventory

Total capital tied up in purchased-but-unsold inventory amounts to $2.71M — funds unavailable for redeployment into higher-performing areas of the business.

Risk Assessment
The $2.71M figure understates total exposure when storage, handling, and markdown costs are factored in. Concentration in a small number of vendor lines amplifies the risk.

Recommended Action
Identify the top 5 vendors contributing most to unsold stock. Initiate clearance promotions for SKUs with StockTurnover < 0.5. Targeting a 30% clearance rate within Q3 could free approximately $810K in working capital.

Finding 6 — The Profitability Paradox

Statistical analysis reveals that low-volume vendors consistently achieve significantly higher profit margins than top-revenue vendors — a counterintuitive result that directly challenges conventional sales-focused procurement thinking.

Vendor Tier	95% Confidence Interval	Mean Profit Margin
Top Performers (top 25% by sales)	30.74% — 31.61%	~31.2%
Low Performers (bottom 25% by sales)	40.48% — 42.62%	~41.5%

The likely drivers of this paradox include: competitive pricing pressure forcing margin compression on high-volume vendors, bulk shipping cost absorptions negotiated into top-vendor contracts, and aggressive retailer discount agreements. Low-volume vendors, by contrast, serve premium niche categories with less pricing competition.

Statistical Validation — Hypothesis Test

Null Hypothesis (H_0)	No significant difference exists in mean profit margins between top and low-performing vendors.
Alt. Hypothesis (H_1)	A statistically significant difference exists in mean profit margins between the two tiers.
Test Used	Two-sample independent T-test (Welch's) — assumes unequal variance between groups
p-value	**0.003**
Decision	**REJECT H_0 — The difference is statistically significant and not attributable to chance**

Conclusion	Low-performing vendors by sales volume demonstrably operate at higher profit margins. The two tiers require distinct management strategies.

04 POWER BI DASHBOARD

An interactive Power BI dashboard was developed to present findings to business and operational stakeholders. Two slicers — Vendor Name (dropdown) and Target Brand Yes/No (buttons) — enable cross-filtering of all nine visual components simultaneously.

Visual Component	Business Purpose
KPI Cards (5)	At-a-glance view of Total Sales, Purchases, Gross Profit, Margin %, and Unsold Capital
Donut Chart — Purchase Contribution %	Visualises top-10 vendor procurement dependency with 65.7% centre annotation
Bar Chart — Top Vendors by Sales	Ranks top 10 vendors by total sales revenue
Bar Chart — Top Brands by Sales	Identifies highest revenue-generating product brands
Bar Chart — Low Performing Vendors	Highlights vendors with the lowest stock turnover ratios
Scatter Plot — Brand Profitability	Maps all brands by Total Sales vs. Avg. Profit Margin with 65% reference threshold line
Key Insights Text Box	Plain-language summary of top 4 findings for non-technical stakeholders
Slicer — Vendor Name	Filters all visuals by a selected vendor
Slicer — Target Brand (Yes / No)	Isolates high-margin, low-sales brands across all visuals

05 STRATEGIC RECOMMENDATIONS

Priority	Action	Estimated Impact
High	Launch promotions for high-margin, low-sales Target Brands	~$4.2M revenue uplift at 20% sales increase
High	Diversify vendor base — reduce top-10 share from 65.7% to <55%	Cuts single-vendor disruption risk by ~40%
High	Clear $2.71M unsold inventory via discounts and bundling	~$810K freed at 30% clearance rate by Q3

Medium	Shift fast-moving SKUs to large-order purchasing thresholds	~$2.1M annual procurement cost savings
Medium	Flag StockTurnover < 0.5 products for quarterly review	Prevents future inventory capital lock-up
Low	Investigate margin compression in top-revenue vendors	Identifies selective price adjustment opportunities
Low	Establish monthly KPI dashboard review cadence	Tracks progress on all initiatives above

06 LIMITATIONS & ASSUMPTIONS

- Analysis covers a fixed time period — seasonal demand fluctuations are not accounted for
- Records with negative profit margins were excluded for analytical clarity — actual loss-making transactions may be understated
- Overhead, warehousing, and logistics costs beyond FreightCost are not incorporated into margin calculations
- Vendor performance tiers use quantile-based thresholds that may shift with different time windows
- Target brand thresholds (15th percentile sales, 85th percentile margin) should be recalibrated periodically

07 CONCLUSION

This end-to-end vendor performance analysis demonstrates a complete, production-grade data analytics workflow applied to real business challenges. The project moved from raw multi-million-row datasets through SQL engineering, Python statistical analysis, and Power BI visualisation to arrive at evidence-based, quantified recommendations.

The Profitability Paradox — confirmed at $p = 0.003$ — is the most strategically significant finding. High-revenue vendors are not the highest-margin vendors. This directly challenges the assumption that maximising sales volume is synonymous with maximising profitability, and provides a clear mandate for margin-aware vendor management.

Combined with $2.71M in locked inventory capital, 65.7% procurement concentration risk, and a portfolio of high-margin brands awaiting promotional activation, this analysis provides a clear, quantified roadmap for improving operational efficiency and unlocking significant untapped revenue potential.

08 TOOLS & TECHNOLOGIES

Category	Technology	Application
Database	SQLite + SQLAlchemy	Raw data storage and multi-table querying
Data Processing	Python + Pandas + NumPy	Cleaning, transformation, feature engineering
SQL	CTEs + JOINs + Aggregations	vendor_sales_summary master table creation
Visualisation	Matplotlib + Seaborn	EDA charts: histograms, boxplots, scatter, heatmap
Statistics	SciPy (ttest_ind, t.ppf)	T-test and 95% confidence interval analysis
BI Dashboard	Power BI Desktop + DAX	Interactive stakeholder-facing dashboard
Version Control	GitHub	github.com/Anujadhav23

— End of Report —